

HOMEWORK 2

>>NAME HERE<<
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Instructions:

Use this LaTeX file as a template to develop your homework and submit it as a single PDF file on Gradescope. For the programming component, you can use any programming language, although the teaching staff will be most able to assist with Python and specifically the package `scikit-learn`; you can use any functionality of the latter. Put all code used for the assignment at the end of the document in whatever way is easiest but still legible.

1 Linear Regression and Regularization

Suppose we model the data-generation process as $y = w^T x + \epsilon$, where inputs x and an unknown parameter w are d -dimensional and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ for some $\sigma \geq 0$.

1.1 Conditional probability [5 points]

Write down an expression for $P(y|x)$.

$$P(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-w^T x)^2}{2\sigma^2}}.$$

1.2 Conditional log-likelihood [10 points]

Assume you are given a set of training observations $(x^{(i)}, y^{(i)})$ for $i = 1, \dots, n$. Derive the conditional log likelihood of this training data. Drop any constants that do not depend on the parameter w .

$$\begin{aligned} P(Y|X) &= \prod_{i=1}^n P(y^{(i)}|x^{(i)}) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y^{(i)}-w^T x^{(i)})^2}{2\sigma^2}} \end{aligned}$$

$$\begin{aligned} \ell(w) &= \log \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y^{(i)}-w^T x^{(i)})^2}{2\sigma^2}} \right) \\ &= \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y^{(i)}-w^T x^{(i)})^2}{2\sigma^2}} \right) \\ &= \sum_{i=1}^n \left(\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) + \log \left(e^{-\frac{(y^{(i)}-w^T x^{(i)})^2}{2\sigma^2}} \right) \right) \\ &= \sum_{i=1}^n \left(\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{(y^{(i)}-w^T x^{(i)})^2}{2\sigma^2} \right) \end{aligned}$$

If we drop the constants that do not depend on w , we get:

$$\ell(w) = - \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2$$

1.3 MLE is the least-squares minimizer [5 points]

Based on your answer above, show that finding the MLE of that conditional log likelihood is equivalent to minimizing least squares, $\frac{1}{2} \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2$.

$$\begin{aligned} w_{MLE} = \arg \max_w \ell(w) &= \arg \max_w - \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2 = \arg \min_w \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2 \\ &= \arg \min_w \frac{1}{2} \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2 \end{aligned}$$

1.4 Ridge regression optimization [5 points]

Suppose we add a Ridge penalty to the objective, i.e. $\frac{1}{2} \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2 + \frac{\lambda}{2} \|w\|_2^2$ for some $\lambda > 0$. Although there is a closed form solution to this problem, there are situations in practice where we solve this via gradient descent. Suppose at the t th step of gradient descent we are at position $w_t \in \mathbb{R}^d$, and we are using a fixed learning rate $\eta > 0$. Compute the position $w_{t+1} \in \mathbb{R}^d$ at the $(t+1)$ st step.

From the gradient $\nabla_w \ell(w_t) = - \sum_{i=1}^n (y^{(i)} - w_t^T x^{(i)}) x^{(i)} + \lambda w_t$ we have the following update:

$$w_{t+1} = w_t + \eta \sum_{i=1}^n (y^{(i)} - w_t^T x^{(i)}) x^{(i)} - \eta \lambda w_t$$

1.5 Effect of the penalty term on optimization [5 points]

Suppose the data is entirely uninformative, i.e. $x^{(i)} = 0_d \forall i$, but we still run gradient descent starting from a nonzero initialization. What is the difference in the behavior of gradient descent with and without the Ridge penalty? Assume $\lambda \leq 1/\eta$.

With the Ridge penalty gradient descent will have the update $w_{t+1} = (1 - \eta\lambda)w_t$, so it will converge to the zero vector. Without the penalty gradient descent will have update $w_{t+1} = w_t$, so it will remain fixed at initialization.

1.6 Bayesian view of Ridge regression [15 points]

Assume a prior on w that each of its d entries is drawn i.i.d. from $\mathcal{N}(0, \tau^2)$. Prove that the MAP estimate of w with this prior is equivalent to minimizing the above regularized least squares problem with $\lambda = \frac{\sigma^2}{\tau^2}$.

To begin, note that $P(w) = \left(\frac{1}{\sqrt{2\pi\tau^2}}\right)^d e^{-\frac{\|w\|_2^2}{2\tau^2}}$ and $P(Y|X, w) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n e^{-\frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2}}$. Thus we have $P(Y|X) = P(Y|X, w)P(w) = \left(\frac{1}{\sqrt{2\pi\tau^2}}\right)^d \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n e^{-\frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2} - \frac{\|w\|_2^2}{2\tau^2}}$ and so

$$\log P(Y|X) = d \log \frac{1}{\sqrt{2\pi\tau^2}} + n \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2} - \frac{\|w\|_2^2}{2\tau^2}$$

If we drop terms that do not depend on w , we get $\ell(w) = -\frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2} - \frac{\|w\|_2^2}{2\tau^2}$. Therefore

$$\begin{aligned} w_{MAP} &= \arg \max_w \ell(w) \\ &= \arg \max_w -\frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2} - \frac{\|w\|_2^2}{2\tau^2} \\ &= \arg \min_w \frac{\sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2}{2\sigma^2} + \frac{\|w\|_2^2}{2\tau^2} \\ &= \arg \min_w \frac{1}{2} \sum_{i=1}^n (y^{(i)} - w^T x^{(i)})^2 + \frac{\sigma^2}{2\tau^2} \|w\|_2^2 \end{aligned}$$

2 Logistic regression with different representations

2.1 Bag-of-words [5 points]

The files `imdb_train.csv` and `imdb_test.csv` both consist of 25K lines, on each of which there is a label (`pos` or `neg`) followed by a tab and followed by a movie review. We will construct Bag-of-words (BoW) representations as follows:

1. **Tokenize the documents** by lower-casing it, replacing all punctuation (e.g. all characters in Python's `string.punctuation`) with spaces, and replacing document by lists of tokens by splitting along spaces.
2. **Construct a vocabulary** by taking the 10K tokens that appear most often in the training data lists.
3. **Represent documents as BoWs** by assigning each token to an index and representing each document in both the training and testing set by a vector $x \in \{0, 1\}^{10K}$ s.t. $x_{[j]} = 1$ if the document contains the j th token and otherwise set $x_{[j]} = 0$.

Represent the training data as the pair $(X_{\text{train}}, y_{\text{train}}) \in \{0, 1\}^{25K \times 10K} \times \{0, 1\}^{25K}$ consisting of the input data X_{train} and labels y_{train} , and similarly represent the test data as the pair $(X_{\text{test}}, y_{\text{test}}) \in \{0, 1\}^{25K \times 10K} \times \{0, 1\}^{25K}$. Train (unregularized) logistic regression on this training data and report the accuracy of the resulting classifier on the test data. Hint: if you use `scikit-learn`'s `LogisticRegression`, setting `solver='liblinear'` tends to lead to consistent convergence.

The test accuracy should be around 0.8414. If using `scikit-learn`, the classifier call should include the kwarg `C=float('inf')` to make it unregularized.

2.2 Standardization [5 points]

Often we standardize the features by applying the transformation $(x_{[j]} - \mu_j)/\sigma_j$ to the j th entry of each input x , where μ_j and σ_j are the empirical mean and standard deviation of the j th feature across the training data. What is a good reason to avoid doing that here?

BoWs are sparse since most tokens do not appear in most documents, and so the matrix X_{train} can be stored with relatively little memory. Unless a token appears in every document, subtracting out the mean causes its entry to not be sparse, making them matrix X_{train} require $10K \times 25K \times 64 \text{ bits} \approx 2\text{Gb}$ RAM to store and process.

2.3 Regularization and cross-validation [5 points]

We can do better by regularizing the norm of the linear classifier using the penalty $\frac{1}{Cn}\|w\|_2^2$ on the classification weights w , where C is an inverse regularization strength and $n = 25K$ is the number of training examples.¹ Train ℓ_2^2 -regularized logistic regression on the training data, with the hyperparameter C selected by 4-fold cross-validation from a set of ten candidate values spaced logarithmically in the interval $[1\text{E-}4, 1\text{E}4]$, and report the accuracy of the resulting classifier on the test data. Hint: you may use `scikit-learn`'s `LogisticRegressionCV`, which defaults to the above candidate values of C .

The test accuracy should be around 0.88448.

2.4 Feature selection [5 points]

Report the value that the linear classifier weight assigns to the tokens “the”, “good”, and “bad” in both the unregularized and the regularized case. What does this say about the effect of regularization?

token	unregularized	regularized
the	3.6297395645859094	0.04199206809227908
good	6.598037200798704	0.1672232691346698
bad	-12.584033784688273	-0.660459054763

While regularization reduces the magnitudes of all values, it reduces the weight assigned to the stopword “the” relatively more than it does the weights assigned to sentiment-relevant content words like “good” and “bad”, thus placing more emphasis on words that we might associate with being in good or bad reviews.

¹We use inverse regularization strength C instead of the usual regularization strength λ because that is how `scikit-learn` does it.

2.5 Latent Semantic Analysis [5 points]

The file `imdb_unsup.txt` contains 50K lines, on each of which there is an unlabeled movie review. Tokenize it and construct BoWs from it as in Question 2.1 to get a matrix $X \in \{0, 1\}^{50K \times 10K}$. To use this data, we will treat tokens as samples and whether or not they appear in document i as features, and obtain a 10-dimensional representation of each token j in the vocabulary by taking the first ten singular values Σ_k and singular vectors U_k of X^T and multiplying them together. This is called Latent Semantic Analysis (LSA) and is a dimensionality reduction alternative to PCA often used for documents. Compute $R^{(LSA)} = U_k \Sigma_k \in \mathbb{R}^{10K \times 10}$ and **briefly give one practical reason why PCA might not be appropriate to use here**. Hint: `scikit-learn`'s `TruncatedSVD.fit_transform` will compute the LSA embeddings directly; if using it, setting `algorithm='arpack'` derandomizes the algorithm and yields more consistent results.

PCA assumes the data is centered, which similarly to Question 2.2 is impractical to do here because subtracting out the means from X will yield a $50K \times 10K$ dense matrix, which requires ≈ 4 Gb RAM to store and process.

2.6 Continuous BoW [5 points]

We now have a low-dimensional representation for each token, but to classify documents with logistic regression we need to represent documents instead. A simple way to do this is to just represent document i by the sum of the representations $R_{[j]}^{(LSA)}$ of all tokens $j \in [10K]$ that occur in that document. This is called a Continuous Bag-of-Words (CBoW) representation. Train and test 4-fold cross-validated logistic regression when the input matrices are the LSA-based CBoW alternatives $X_{\text{train}}^{(LSA)}, X_{\text{test}}^{(LSA)} \in \mathbb{R}^{25K \times 10}$ (the label vectors $y_{\text{train}}, y_{\text{test}}$ remain the same) and report the test accuracy.

The test accuracy should be around 0.78804.

2.7 Advantages and disadvantages of dimensionality reduction [5 points]

Ignoring accuracy, list one advantage and one disadvantage of using these LSA-based CBoW representation.

An advantage is that the CBoW dimension is ten, which is much smaller than the BoW dimension or even the average sparsity, making storage and processing more efficient. A disadvantage is that looking at the logistic regression coefficients tells us nothing about the weights we assign to individual words, reducing interpretability.

2.8 Choice of embedding dimension [5 points]

We specified you use an embedding dimension of 10 for LSA, but list two reasonable ways you could have chosen a suitable dimension using the data. Hint: one way uses the supervised training data, and another way uses only the unsupervised data.

1. Using cross-validation on the IMDb task, just like we are doing to pick the regularization strength.
2. Looking at the explained variance and finding a “knee.”

2.9 You shall know a word by the company it keeps (Firth, 1957) [5 points]

The file `glove.csv` contains a token, followed by a tab, followed by a space-delimited list of 300 numbers representing that token. This is a subsampled set of GloVe word vectors (<https://nlp.stanford.edu/projects/glove/>) trained on 840B tokens of Common Crawl web data; these vectors are produced using a different unsupervised model that counts documents as similar if they co-occur within a small span of words of each other. Use these GloVe vectors to construct input matrices $X_{\text{train}}^{(GloVe)}, X_{\text{test}}^{(GloVe)} \in \mathbb{R}^{25K \times 300}$ consisting of 300-dimensional GloVe-based CBoWs, train and test 4-fold cross-validated logistic regression on them, and report the test accuracy.

The test accuracy should be around 0.85664.

2.10 Learning curves [5 points]

Train 4-fold cross-validated logistic regression on BoW, LSA-based CBoW, and GloVe-based CBoW document representations while varying the amount of training data provided, specifically using the following number of training examples: 8, 40, 200, 1K, 5K, and 25K. Make the number of positive and negative samples the same in all datasets and plot the test accuracy vs. number of examples of all three approaches.

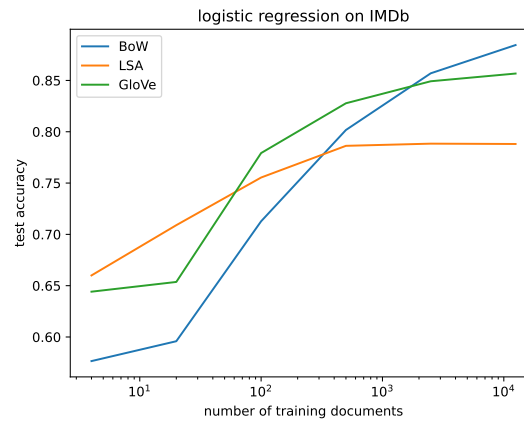


Figure 1: Question 2.10

See Figure 1. The position of the beginning of the curve can vary based on randomly selected subsamples, but the two unsupervised approaches should do better the beginning and the supervised-only one should do better at the end.

2.11 All you need is GloVe [5 points]

Discuss what methods perform best at different accuracy levels. What regimes is unsupervised learning most useful in?

As shown in Figure 1, in-domain unsupervised learning (i.e. LSA-based CBoW) performs best at very few samples (8-40), general-purpose unsupervised learning (GloVe) performs well in the middle range (40-5K), and using labeled data alone performs best at larger sample sizes (5K-25K). This highlights the usefulness of unsupervised learning on downstream supervised learning tasks when the number of labeled examples is small.